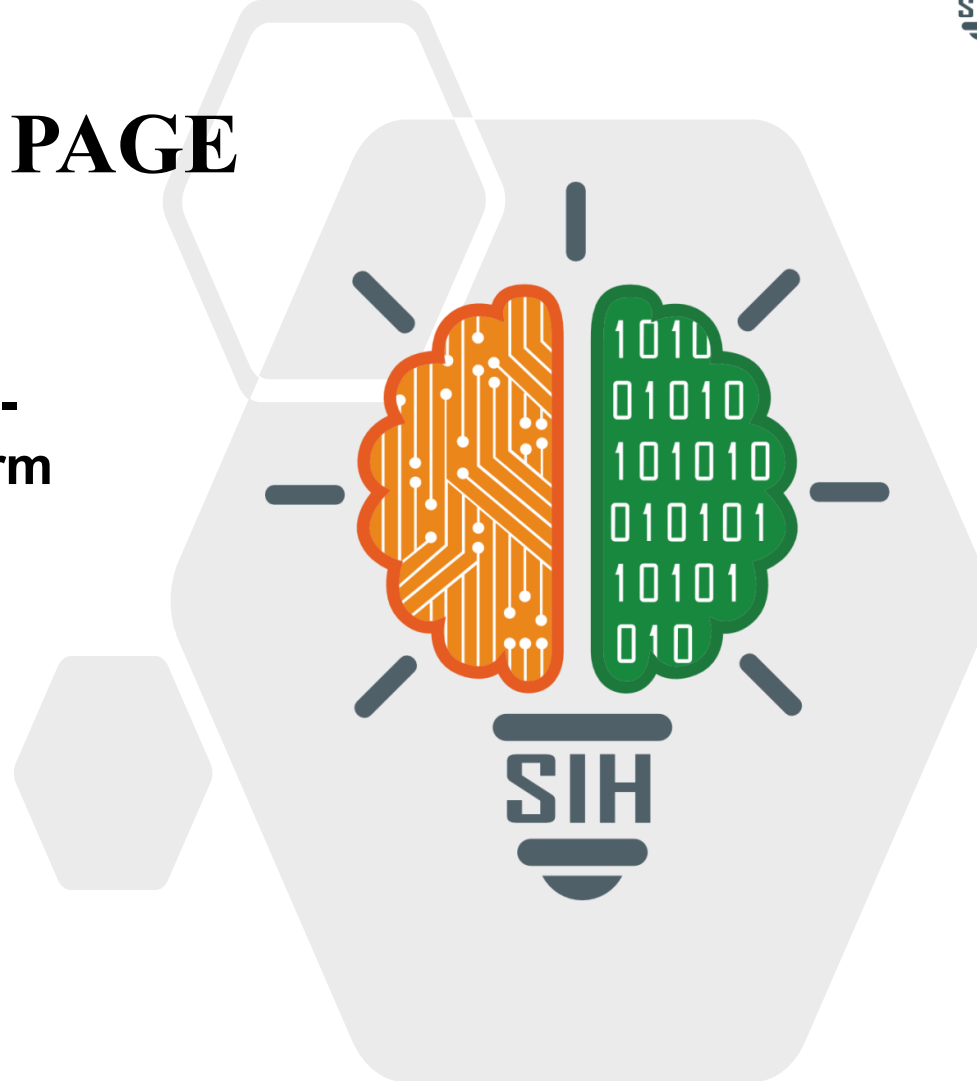


TITLE PAGE

- Problem Statement ID - SIH26067
- Problem Statement Title - Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.
- Theme - Disaster Management
- PS Category - Software
- Team ID - K26125
- Team Name - Team Null





VWater: Ocean in 3D



Forecasters get the ocean model and the measurements in separate tools, as flat maps. **VWater puts both in one 3D scene in the browser, and you can ask it questions or fly over it.**

THE PROBLEM

The ocean is 3D; the tools are flat

A cyclone feeds on a warm layer tens of metres thick. A surface map cannot show how thick.

Model and measurements live apart

The model in one program, float and glider files in another; comparing them is done by eye.

WHAT WE BUILT

One link, no install

any browser; deployable on INCOIS servers

Model and measurements in one 3D scene

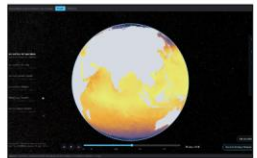
the INCOIS model with 35 Argo float profiles and 112 glider casts

Click a float, see the gap

its profile against the model, in °C and in cyclone heat potential

Bad readings stay visible

failed quality checks drawn red, never dropped



Simple
the whole ocean



Region 3D
the water column



Map 2D
any depth

WHAT IS NEW

1 · Ask the ocean in plain words

Assistant

Ask me about the ocean data on screen, or about any control or word in the viewer. Numbers I give come from the data; anything I cannot trace back is marked.

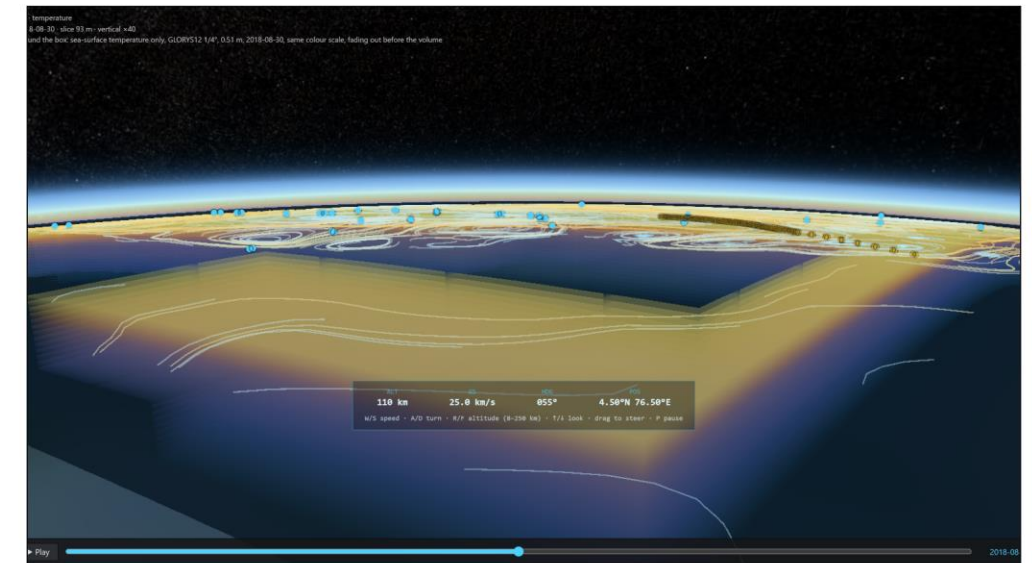
Open float 2902596 and tell me how its cyclone heat potential compares with the model.

The float 2902596 profile is now open. Its cyclone-heat potential (TCHP) is 36.2 kJ cm^{-2} , while the INCOIS analysis gives 50.28 kJ cm^{-2} – the model is about 14 kJ cm^{-2} higher. The depth where the water cools to 26°C (D26) is 42.9 m in the observations versus 63.5 m in the model. The measurement is a delayed-mode (D) cast, QC-accepted flags 1, 2, 5, 8, from the file argo_indian_ocean_20180820.nc.

from: profile, ui action
changed: open profile

Answers from the data and drives the viewer. Here it opened the float, drew its profile and compared it with the model. Any number it cannot trace to the data is flagged.

2 · Fly over the Bay like a plane



A flight view with pilot controls: speed, turn, altitude 8–250 km. The 3D water column, the currents and every float pass underneath at 40× depth.

Why it matters

a 2,000 m-deep ocean becomes something a student, a forecaster or a minister can see and move through, not a stack of flat maps

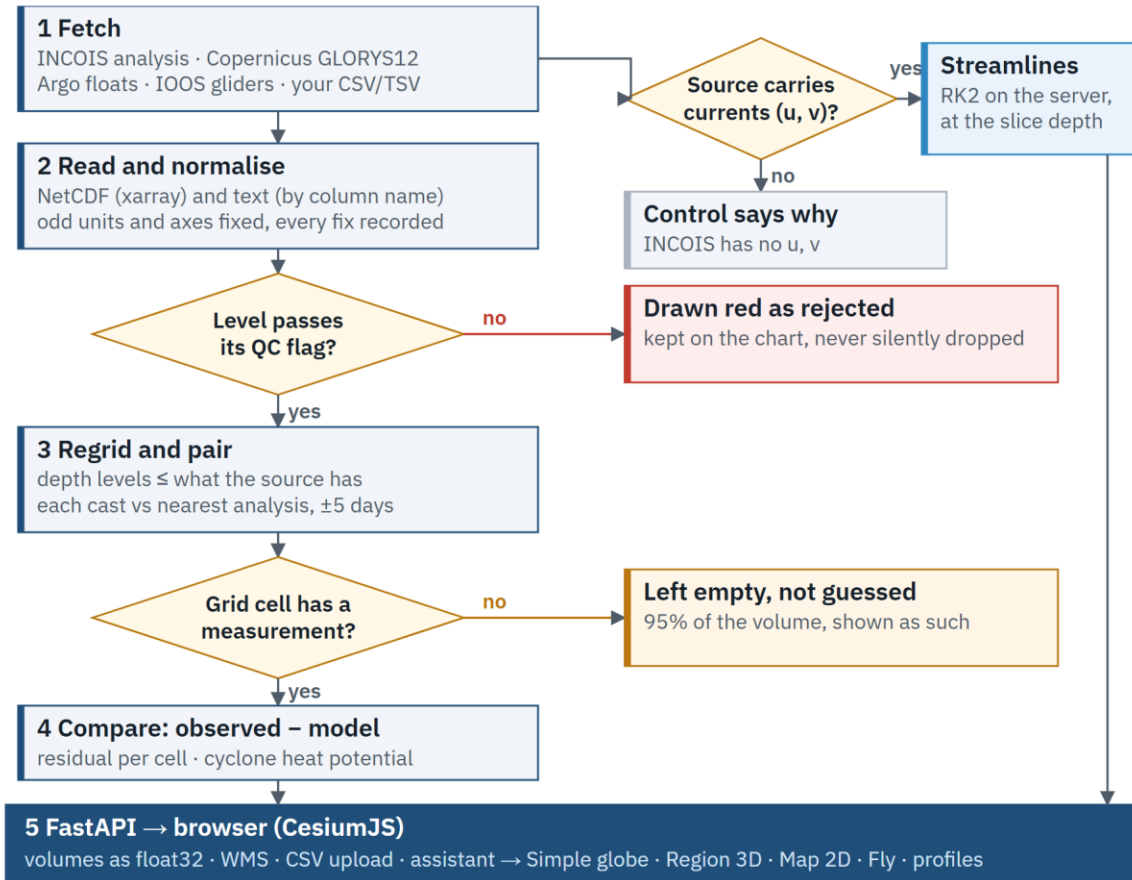
How to fly

W/S speed · A/D turn · R/F altitude · drag to steer



TECHNICAL APPROACH

FROM RAW FILES TO THE SCENE — AND WHAT HAPPENS TO BAD OR MISSING DATA



Diamonds are decisions the code makes. The red and amber branches are where most tools quietly drop a bad reading or paint over a gap; here both stay on screen, labelled.

WHAT WE USED, AND WHY

CesiumJS

a real 3D globe with a built-in volume renderer

TypeScript + Vite

the viewer; no heavy UI framework

Python: FastAPI + xarray

reads the NetCDF and text files the brief names

Raw float32 volumes

one full 3D timestep is 0.55 MB, straight to the GPU

OGC WMS

the same layers open in QGIS or any national portal

Groq LLM + our own tools

the assistant can only answer through our data API



Step 4 on screen: each block is a place someone measured, coloured by measured minus model (red warmer, blue colder). Unmeasured water stays empty.



FEASIBILITY AND VIABILITY



35 + 112

Argo float profiles and glider casts, paired with the model

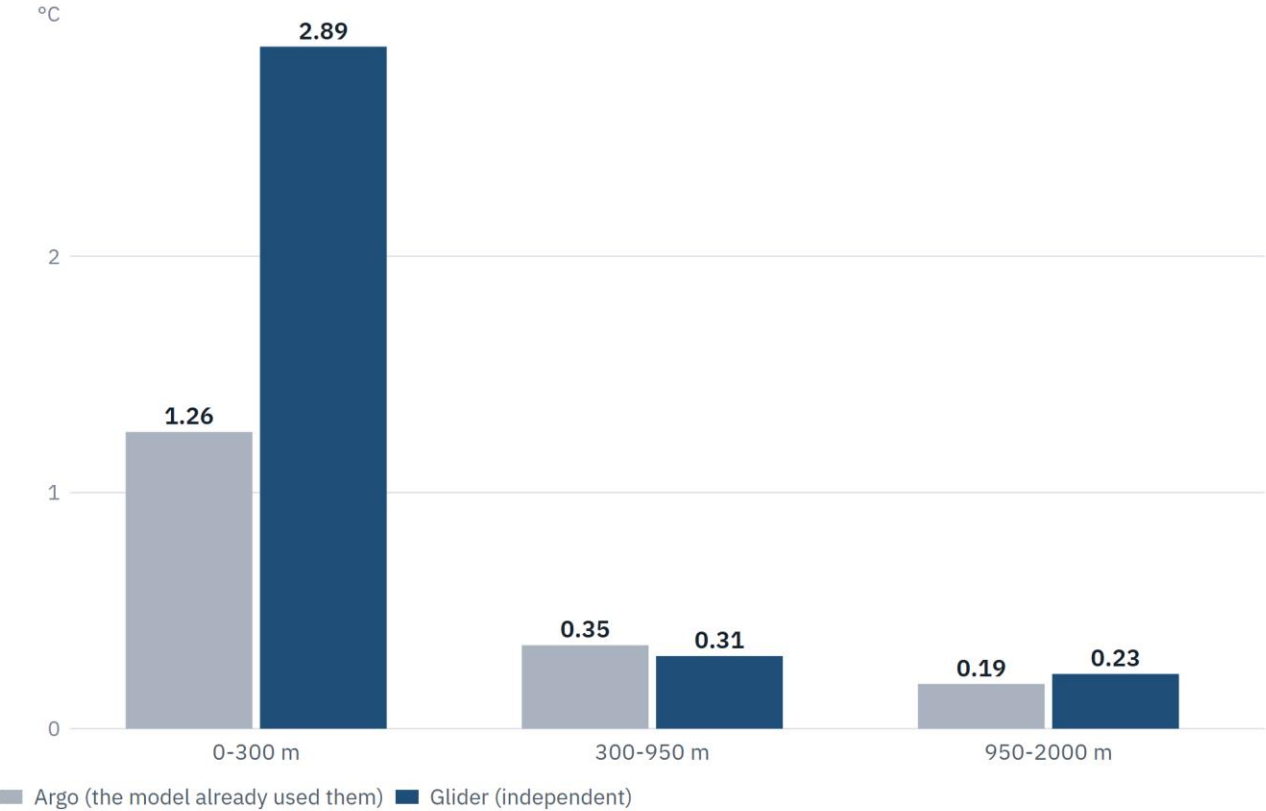
0.55 MB

one full 3D timestep, as sent to the browser

0.45 s

to compare 127 casts with the model and build the 3D error layer

MEASURED FINDING: MODEL ERROR BY DEPTH



Below 300 m the model and the instruments agree. In the top 300 m — the warm layer a cyclone feeds on — the independent glider finds an error of **2.89 °C**. VVater makes that gap visible.

RISKS WE HIT, AND WHAT WE DID

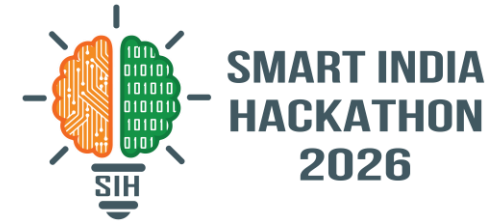
RISK	HANDLED BY
Brief's data links are dead	Both are ftp: // and blocked; replaced with tested HTTPS mirrors.
Bad float readings	510 of 11,525 levels fail QC; drawn red, never dropped.
Files not quite standard	Odd units and axes read defensively; every fix recorded with the data.
Unknown GPUs at INCOIS	Four quality tiers, picked by measuring real frames at start-up.
Copernicus needs an account	INCOIS data needs none and is the default; Copernicus only adds currents.

Runs on what INCOIS has

One Python process and static files. Any modern browser, no install, no account.



IMPACT AND BENEFITS



FOR THE CYCLONE FORECASTER

+16.0

kJ/cm^2 — model minus measured **cyclone heat potential** along the independent glider track (95 casts, one deployment; measured mean 9.5). Against Argo, which the model already absorbs: -3.5 .

Fuel, in the unit already used

heat above 26°C , per float and per glider cast

20°C isotherm in one click

the standard proxy for how deep the warm layer goes

Model and truth side by side

no second program, no copying numbers between tools

Confidence visible

uncertain water drawn faint; unmeasured water drawn empty

FOR INCOIS OPERATIONS

Search and rescue, fisheries, climate

currents and temperature at any depth, in the same view

New instruments without re-engineering

drop a CSV of casts on the globe; it is paired with the model at once

Open standards

WMS layers open in QGIS and national portals

FOR STUDENTS, PUBLIC AND POLICY

4.86%

of the Bay's water column had a measurement in this window. Shown honestly, that is itself the case for more floats — a sentence a policymaker can repeat.

True scale in one click

the Bay really is a film of water: 2 km deep, 2,400 km wide. Then $40\times$, to read it

From planet to profile

a simple globe, then the Bay in 3D, then fly over it, then one float's dive

Runs in a school's browser

exhibitions, e-learning and outreach, as the brief asks

Ask it in plain words

a built-in assistant answers from the data and moves the view for you

WHERE IT GOES NEXT

More regions and years

the region and window are configuration, not code

More sensors

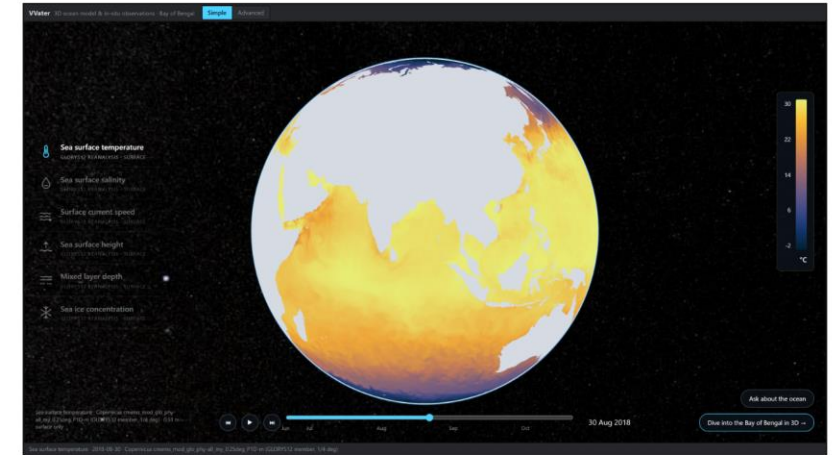
CTD, moorings, HF radar, ADCP: each one parser

Machine-learning products

any gridded output renders as another volume



Where the model is wrong. Each block is a place someone measured, coloured by measured minus model (red warmer, blue colder). Everything else is empty on purpose: 95% of the volume.



The Simple view, for everyone else. The world's ocean one layer at a time — temperature, salinity, currents, sea level, mixed layer, ice — on a timeline.



RESEARCH AND REFERENCES



Source code, docs and every measured number: <https://github.com/Kukyos/VVater>

public repository · all links below opened 2026-09-23

DATA WE FETCHED — EACH ONE PROBED, NOT ASSUMED

- 1 INCOIS ERDDAP — Argo 10-day variational analysis ([incois_argo_10d_VAM](https://erddap.incois.gov.in/erddap/griddap/incois_argo_10d_VAM.html)), temperature and salinity with per-cell error. The primary volume; no credentials.
erddap.incois.gov.in/erddap/griddap/incois_argo_10d_VAM.html
- 2 Copernicus Marine — GLORYS12 global reanalysis, GLOBAL_MULTIYEAR_PHY_001_030. Currents, and the global surface layer.
data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_001_030
- 3 Argo Global Data Assembly Centre, HTTPS mirror. Float profiles with QC flags and data mode.
data-argo.ifremer.fr
- 4 U.S. IOOS Glider DAC — Rutgers *ru29* deployment, NetCDF and CSV. The independent check.
gliders.ioos.us/erddap
- 5 INCOIS Live Access Server. Named in the brief; reachable, not yet ingested.
las.incois.gov.in/las
- 6 The brief's two [ftp.ifremer.fr](ftp://ftp.ifremer.fr/ifremer/argo) links. Probed: do not connect (port 21). Replaced by 3.
<ftp://ftp.ifremer.fr/ifremer/argo> · [.../glider/v2](https://glider.v2)
- 7 Argo Program overview, Scripps Institution of Oceanography. What a float is and does.
argo.ucsd.edu
- 8 Smart India Hackathon 2026, problem statement 26067 (MoES / INCOIS Ocean Valley). Quoted verbatim in the repository.
github.com/Kukyos/VVater/blob/main/docs/01-problem-statement.md

How these links were checked

Every URL was opened on 2026-09-23 with a browser user agent and returned HTTP 200, except the two FTP links (dead, listed as such) and the AMS journal page, which refuses scripted requests; its DOI resolves.

STANDARDS AND METHODS THE CODE FOLLOWS

- 9 Wong et al., Argo Quality Control Manual for CTD and Trajectory Data. Range test and QC flags.
doi.org/10.13155/33951
- 10 U.S. IOOS QARTOD manuals. Glider QC flags.
ioos.noaa.gov/project/qartod
- 11 NetCDF Climate and Forecast (CF) Conventions. Read defensively; assumptions recorded.
cfconventions.org
- 12 OGC Web Map Service 1.3.0. The WMS endpoint.
ogc.org/standards/wms
- 13 IOC, SCOR, IAPSO (2010), TEOS-10 seawater equation. Pressure to depth, heat capacity.
teos-10.org
- 14 Leipper & Volgenau (1972), Hurricane heat potential of the Gulf of Mexico, J. Phys. Oceanogr. 2. Heat above 26 °C.
[doi.org/10.1175/1520-0485\(1972\)002<0218:HHPOTG>2.0.CO;2](https://doi.org/10.1175/1520-0485(1972)002<0218:HHPOTG>2.0.CO;2)
- 15 Thyng et al. (2016), True colors of oceanography, Oceanography 29(3). cmocyan palettes.
doi.org/10.5670/oceanog.2016.66
- 16 CesiumJS VoxelPrimitive reference. The volume renderer (experimental, pinned).
cesium.com/learn/cesiumjs/ref-doc/VoxelPrimitive.html
- 17 xarray and FastAPI documentation. Ingest and API.
docs.xarray.dev · fastapi.tiangolo.com
- 18 Unidata NetCDF. The file format the brief names.
unidata.ucar.edu/software/netcdf